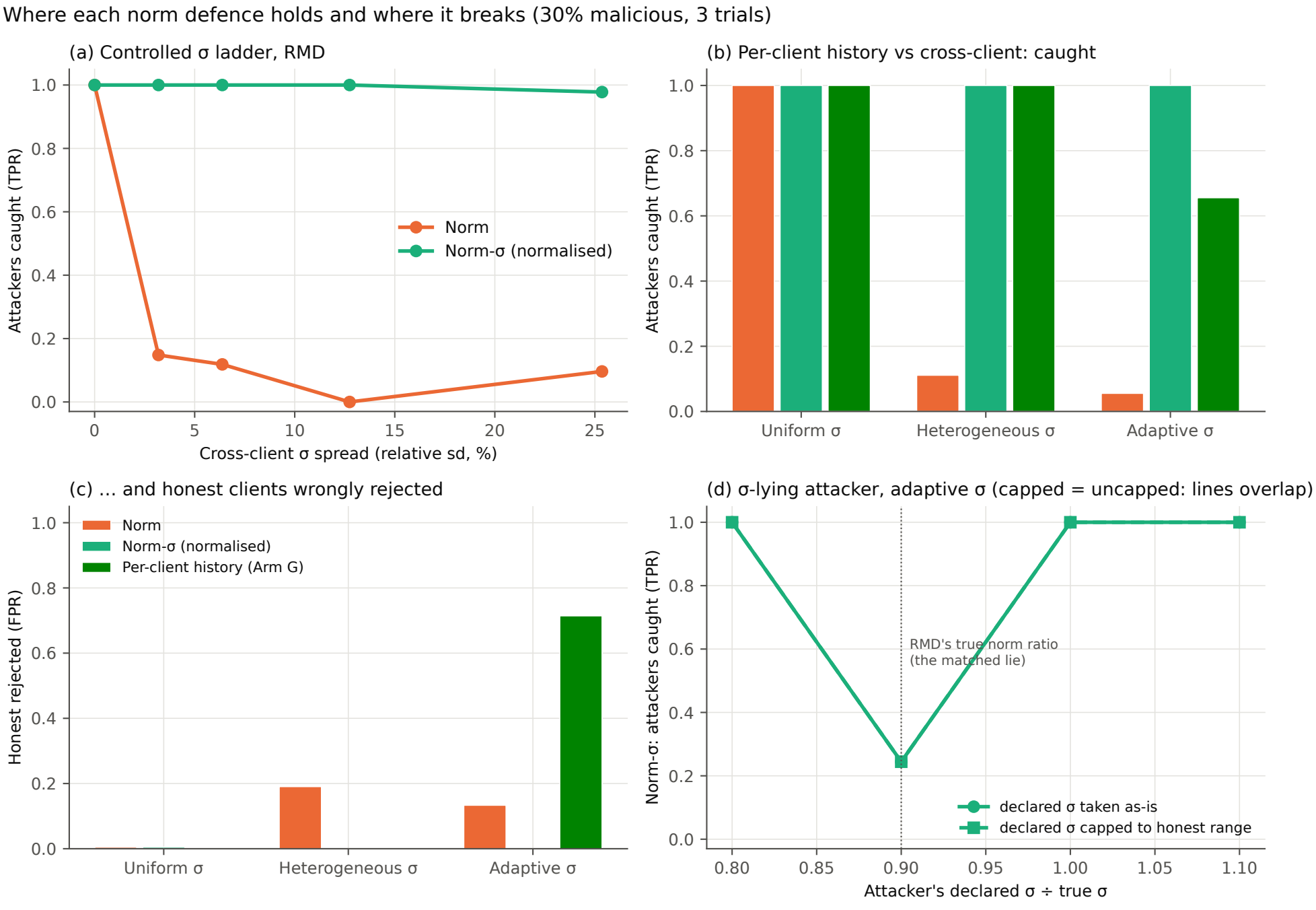
Where each norm defence holds and where it breaks (30% malicious, 3 trials)
(a) Controlled σ ladder, RMD
Attackers caught (TPR)
Cross-client σ spread (relative sd, %)
Norm
Norm-σ (normalised)
(b) Per-client history vs cross-client: caught
Attackers caught (TPR)
Uniform σ
Heterogeneous σ
Adaptive σ
(c) ... and honest clients wrongly rejected
Honest rejected (FPR)
Norm
Norm-σ (normalised)
Per-client history (Arm G)
Uniform σ
Heterogeneous σ
Adaptive σ
(d) σ-lying attacker, adaptive σ (capped = uncapped: lines overlap)
Norm-σ: attackers caught (TPR)
RMD's true norm ratio
(the matched lie)
declared σ taken as-is
declared σ capped to honest range
Attacker's declared σ ÷ true σ